This ms aims to investigate the effect of a static electric field on particle trapping due to the wave-particle interaction with a circularly polarized electromagnetic wave. At the present stage of development of plasma science, a thorough exploration of this situation, based on new results and insights previously developed by many researchers, would be a most valuable contribution to the field. Unfortunately, ms PoP26-AR-00477 presents very few new results, actually only the limited numerical results in Fig. 6, found in Appendix B. The ms also fails to connect to the large body of knowledge that underlies the physics being addressed. Based on these shortcomings, it is the opinion of this referee that this ms should not be published in any journal. But, yes, the authors are encouraged to pursue their studies, and in the future deliver a substantial piece of work that moves the frontier understanding on the effect of a DC field on trapping. But such an effort cannot be just a cosmetic repackaging of the present materials, it would need to have real substance.

The presentation of the analysis is awkward, the English is incomplete at times, and the explanation of the figures is poor. The referee can certainly provide suggestions on improvements, but will refrain from doing so because that is not the reason for rejection. It is the lack of physics detail and limited new results, compounded by the disconnect with previous developments, that render this ms not publishable.

The original identification of particle trapping by a circularly polarized EM wave, including the relativistic derivation of the exact pseudo potential (something that the authors of ms PoP26 - AR-00477 attempt in an awkward manner in Sec. II), is given in C. S. Roberts and S. J. Buchsbaum, Phys. Rev. 135, A381 (1964). They outline the trapping conditions and indicate that there are exact solutions in terms of elliptic integrals. T. M. O'Neil in Phys. Fluids 8, 2255 (1965) uses such elliptic integrals to calculate how particle trapping modifies the damping of electrostatic waves, and importantly elucidates the key concept of a trapping frequency scaling as the square root of the electric field of the wave, which more generally is the force exerted by the wave, and is the relevant issue for the topic studied in PoP26-AR-00477. P. Palmadesso and G. Schmidt in Phys. Fluids 14, 1411 (1971) show how the results of trapping by an electrostatic wave map into the analysis of trapping by a circularly polarized EM wave. In this case the trapping frequency instead of being proportional to the square root of the wave electric field, it is proportional to the square root of the wave magnetic force, hence it is proportional to the square root of the perpendicular velocity, which is the reason why the push provided by the DC electric is canceled if the perpendicular velocity grows, as seen in Fig. 6 of PoP26-AR-00477. The details of how trapping by an electrostatic wave is modified by the application of a DC electric field are given by G. J. Morales in Phys. Fluids 23, 2472 (1980). The relevance to PoP26-AR-00477 is that conservation of energy results in growth of the wave amplitude, which, as seeing from the Palmadesso and Schmidt work, is analogous to increasing the perpendicular velocity for the circular polarization case. Thus, as Palmadesso and Schmidt showed the mapping between trapping dynamics in an electrostatic wave to trapping by a circularly polarized wave, a similar mapping of the effect of a DC electric field can be done. An effective way of accomplishing this connection is by adding a DC electric field to Eqs. 3a-4b in R. F. Lutomirski and R. N. Sudan Phys. Rev. 147, 156 (1966). But that is just enough hints for the authors of PoP26-AR-00477 of what is behind what they are trying to investigate. The predicted particle clumping by the combination of wave trapping and a DC electric field has been measured in the laboratory, is used in the operation of travelling wave tubes, and has been considered as method for enhancing the efficiency of free electron lasers.

More can be said about magnetospheric studies considering trapping and DC electric fields, dating back to the work of A. Brinca in Geophys. Res. Lett 7, 901 (1980), but this small survey should be adequate to ascertain that there is a sizeable body of knowledge that should be considered by the authors in a possible future publication on this topic.